

Paper IV - Recent Advances (MCh Urology, Summer 2015)

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IV-1. Role of robotics in urology

Type: LAQ (25 marks)

Introduction / Definition

- A robot is a reprogrammable, multifunctional manipulator. Current surgical systems are **master-slave telemanipulators**: the surgeon controls every movement (not autonomous).
- Urology adopted robotics earlier and more widely than any other specialty, driven by **RARP (robot-assisted radical prostatectomy)**. RARP is now the commonest approach to radical prostatectomy in many high-income countries.

History (milestones)

Year	Milestone
1985	PUMA 560 : first surgical robot (brain biopsy)
Late 1980s	PROBOT (Imperial College, London): first urological robot, for automated prostate resection
2000	da Vinci cleared by the FDA (Food and Drug Administration); Binder & Kramer performed the first robotic radical prostatectomy
Early 2000s	Menon standardised the VIP (Vattikuti Institute Prostatectomy), which popularised RARP

da Vinci system: components

1. **Surgeon console**: 3D high-definition viewer, master controllers and foot pedals, with motion scaling and tremor filtration.
2. **Patient-side cart**: 3–4 robotic arms carrying the camera and **EndoWrist** instruments with 7 DOF (degrees of freedom).
3. **Vision cart**: image processing, light source, energy generator and assistant monitor.
 - **Generations**: S (2006), Si (2009; dual console, Firefly fluorescence), **Xi (2014)**, **SP** (single 2.5 cm port; urology 2018) and **da Vinci 5** (2024; force feedback).
 - **Other platforms**: Hugo RAS, Versius, Senhance and the low-cost Indian **SSI Mantra**.

Advantages and disadvantages

Advantages	Disadvantages
3D magnified vision	High capital cost (~US\$1.5–2.5 million) plus disposables
Wristed instruments with 7 DOF	Limited haptic feedback
Tremor filtration and motion scaling	Docking time and large footprint
Better surgeon ergonomics	Surgeon is remote from the patient (emergency undocking)
Easier intracorporeal suturing than laparoscopy	Learning curve and credentialing issues
Less blood loss and shorter stay than open surgery	Limited availability in the public sector

Applications in urology (with evidence)

1. Radical prostatectomy (index procedure)

- **Yaxley et al., Lancet 2016** (RCT (randomised controlled trial), open vs robotic): urinary and sexual function were **similar at 12 weeks**. The robotic arm had less blood loss and a shorter stay.
- **Coughlin et al., Lancet Oncol 2018** (same trial): function was still **similar at 24 months**.
- Oncological outcomes are comparable to open surgery. Refinements include Retzius-sparing RARP and NeuroSAFE.

2. RAPN (robot-assisted partial nephrectomy): compared with laparoscopy it gives **shorter warm ischaemia** and **less conversion to radical nephrectomy**, and allows complex and hilar tumours to be treated. Firefly ICG (indocyanine green) aids selective clamping.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Parekh, Lancet 2018)**: RARC was **non-inferior** to open cystectomy for **2-year PFS (progression-free survival)**, with less blood loss and fewer transfusions.
- **iROC (Catto, JAMA 2022)**: RARC with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days** and fewer thromboembolic and wound complications.

4. Reconstruction: pyeloplasty (success above 90–95%), ureteric reimplantation, and **robotic BMG (buccal mucosa graft) ureteroplasty** for long ureteric strictures.

5. Other procedures: R-RPLND (robotic retroperitoneal lymph node dissection), adrenalectomy, simple prostatectomy, sacrocolpopexy, and **RAKT (robot-assisted kidney transplantation)**, which reduces wound complications in obese recipients.

Training and credentialing

- Training follows a structured pathway: didactics, then **virtual-reality simulators**, then dry and wet lab, then supervised modular console time.
- **FRS (Fundamentals of Robotic Surgery)** and the **ERUS (EAU Robotic Urology Section)** curriculum; dual consoles allow mentoring.
- RARP learning curve: more than 150–250 cases for optimal margins and continence.

Cost

- Capital, maintenance and limited-use instruments drive cost, so robotics is cost-effective only at **high volume**. Cheaper platforms and reusable instruments may help.

Complications specific to robotics

Problem	Details
Steep Trendelenburg (25-45°)	Raised IOP (intraocular pressure), ischaemic optic neuropathy (rare, causes blindness), corneal abrasion, raised ICP (intracranial pressure), facial and laryngeal oedema
Positioning	Brachial plexus injury; compartment syndrome and rhabdomyolysis after prolonged lithotomy
Instruments	Insulation failure causing stray energy and bowel burns
Lack of haptics	Tissue tearing and suture breakage
Others	Port-site hernia, CO ₂ embolism, system failure (rare)

- Prevention: limit Trendelenburg angle and time, pad well, keep lithotomy under about 2-3 h.

Recent advances

- **Single-port surgery** allows extraperitoneal and transvesical RARP, with possible same-day discharge.
- **Telesurgery**: long-distance robotic operations have been reported from China and India (SSI Mantra) since 2023-25. Latency is the main limitation.
- **AI (artificial intelligence)**: video-based skill assessment, and kinematic data that predict continence after RARP.

Diagram to draw: da Vinci block diagram (console, vision cart, patient cart) and RARP port placement.

Key points to remember:

- PUMA 560 (1985) was the first surgical robot and PROBOT the first urological one; da Vinci was FDA-approved in 2000, the year of the first RARP.
- Yaxley and Coughlin: open and robotic prostatectomy give similar function at 12 weeks and 24 months.
- RAZOR: RARC is non-inferior for 2-year PFS; iROC: intracorporeal RARC gives more days alive and out of hospital.
- Steep Trendelenburg raises IOP and ICP (optic neuropathy); prolonged lithotomy causes compartment syndrome.

IV-2. Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation

Type: LAQ (25 marks)

Introduction

- The organ shortage has led to wider use of **marginal (extended criteria) donors**, whose kidneys carry a higher risk of DGF (delayed graft function), primary non-function or shorter graft survival.

Part A: Marginal kidney donor

Definitions and categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002

- Any donor aged **≥60 years**, or a donor aged **50-59 years** with **at least 2** of: **hypertension, terminal creatinine >1.5 mg/dL, or death from CVA (cerebrovascular accident)**.
- ECD kidneys have about a 70% higher relative risk of graft loss than SCD (standard criteria donor) kidneys.

2. KDRI (Kidney Donor Risk Index) and KDPI (Kidney Donor Profile Index)

- KDRI combines 10 donor factors; KDPI is its percentile. **KDPI >85%** replaced the ECD label in the US (December 2014).

3. DCD (donation after circulatory death): Maastricht classification

Category	Description	Type
I	Dead on arrival	Uncontrolled
II	Unsuccessful resuscitation	Uncontrolled
III	Awaiting cardiac arrest after withdrawal of therapy	Controlled (commonest)
IV	Cardiac arrest in a brain-dead donor	Controlled or uncontrolled
V	Euthanasia	Controlled

- DCD kidneys have **higher DGF (roughly 40-50%)** but **similar long-term graft survival**.

4. Other marginal donors

- AKI (acute kidney injury)** donors: do not discard on terminal creatinine alone.
- HCV (hepatitis C virus)-viraemic donors to HCV-negative recipients**, followed by **DAAs (direct-acting antivirals)**, which cure nearly 100%.
- Marginal living donors**: older or with controlled hypertension; KDIGO (Kidney Disease: Improving Global Outcomes) 2017 advises a **risk-based approach**.

Assessment and optimisation of marginal kidneys

- Remuzzi biopsy score**: glomeruli, tubules, interstitium and arteries are each scored 0-3 (total 0-12).

Score	Action
0-3	Single transplant
4-6	Dual kidney transplant
≥7	Discard

- **HMP (hypothermic machine perfusion):** **Moers et al., NEJM 2009** showed that HMP **reduces DGF (about 21% vs 27%)** and improves 1-year graft survival, with the greatest benefit in ECD kidneys.
- **NMP (normothermic machine perfusion)** at 37 °C allows viability assessment and repair (trials). Keep **cold ischaemia under 12-18 h**.
- **Dual kidney transplant:** two marginal kidneys into one recipient to supply enough nephron mass.
- **Recipient matching:** the **ESP (Eurotransplant Senior Program, 1999)** gives kidneys from donors **≥65 years** to recipients **≥65 years**, prioritising short cold ischaemia over HLA (human leucocyte antigen) matching. Recipients must give **informed consent**.

Outcomes

- **Ojo et al. (JASN 2001):** ECD transplantation improves survival over dialysis, although graft survival is lower than with standard kidneys.
- With DGF, delay nephrotoxic CNI (calcineurin inhibitor) therapy under antibody induction.

Part B: Newer immunosuppressant drugs

Overview of the current standard

- Induction with basiliximab or rATG (rabbit antithymocyte globulin), then **tacrolimus + mycophenolate + steroids**.
- Newer agents aim to avoid CNI nephrotoxicity, improve adherence and treat AMR (antibody-mediated rejection).

Induction agents

Drug	Mechanism	Key points
Basiliximab	Anti-CD25 (IL-2 (interleukin-2) receptor α), non-depleting	20 mg IV (intravenous) on day 0 and day 4 ; for low-risk patients
rATG	Polyclonal T-cell depletion	For high immunological risk or DGF; risks CMV (cytomegalovirus) and PTLD (post-transplant lymphoproliferative disorder)

Maintenance agents (newer and modified)

Drug	Mechanism	Key points
Extended-release tacrolimus (Advagraf, Envarsus XR)	Calcineurin inhibition	Once-daily dosing improves adherence
mTOR (mammalian target of rapamycin) inhibitors: sirolimus, everolimus	Block mTOR and arrest the cell cycle	Not nephrotoxic; fewer CMV/BK infections . TRANSFORM (JASN 2018): everolimus with reduced CNI was non-inferior . Cause poor wound healing and lymphocele
Belatacept	CTLA4-Ig (cytotoxic T-lymphocyte antigen 4-immunoglobulin); blocks CD28 costimulation (signal 2)	FDA (Food and Drug Administration) 2011; 5 mg/kg every 4 weeks for maintenance. BENEFIT/BENEFIT-EXT: better GFR (glomerular filtration rate) than ciclosporin but more early rejection . Contraindicated in EBV (Epstein-Barr virus)-seronegative recipients (PTLD risk)

Desensitisation and antibody-mediated rejection

Drug	Mechanism	Use
IVIG (intravenous immunoglobulin)	Immunomodulation	Desensitisation; AMR with plasmapheresis
Eculizumab	Anti-C5 complement blockade	Prevents aHUS (atypical haemolytic uraemic syndrome) recurrence ; meningococcal vaccination is mandatory
Imlifidase	Enzyme that cleaves IgG within hours	EMA (European Medicines Agency) conditional approval 2020 for highly sensitised patients with a positive crossmatch. Give ATG/rituximab afterwards, as it also cleaves therapeutic IgG

- **Caution about voclosporin:** it is approved for lupus nephritis (2021), **not for renal transplantation**.

Tolerance and the future

- Kidney plus haematopoietic stem-cell transplantation (mixed chimerism) has let selected patients stop immunosuppression.

Side-effect summary

Drug	Signature toxicity
Tacrolimus	Nephrotoxicity, NODAT (new-onset diabetes after transplant), tremor
Mycophenolate	GI (gastrointestinal) upset, teratogenicity
Belatacept	PTLD, early rejection

Recent advances

- Normothermic perfusion with ex vivo repair.
- **Pig-kidney xenotransplantation** into a living human (MGH (Massachusetts General Hospital), March 2024).

Diagram to draw: three-signal T-cell activation model, marking belatacept (signal 2) and basiliximab/mTOR inhibitors (signal 3).

Key points to remember:

- ECD (UNOS 2002): age ≥ 60 , or 50–59 with ≥ 2 of hypertension, creatinine >1.5 mg/dL, CVA death; replaced by KDPI $>85\%$ in December 2014.
- Remuzzi score 0–3 single, 4–6 dual, ≥ 7 discard; HMP reduces DGF (Moers 2009); ESP matches ≥ 65 donors to ≥ 65 recipients.
- Belatacept gives better GFR but is contraindicated in EBV-seronegative recipients; TRANSFORM showed everolimus with low-dose CNI is non-inferior.
- Imlifidase has EMA approval (2020) for highly sensitised patients; voclosporin is not approved for transplantation.

IV-3a. Etiology and management of hypospadias fistula

Type: SAQ (10 marks)

Definition and incidence

- A UCF (urethrocutaneous fistula) is an abnormal communication between the neourethra and penile skin. It is the **commonest complication of hypospadias repair**.
- Incidence: under 5–10% after distal TIP (tubularised incised plate) repair, 10–25% or more after proximal repairs, highest after redo.

Aetiology (risk factors)

- **Distal obstruction** (meatal stenosis, stricture) causes high voiding pressure on the repair. It is the **most important correctable cause**.
- Ischaemia, infection, haematoma and tension impair healing.
- **Overlapping suture lines** (urethral and skin lines on top of each other) and **no barrier layer**.
- Poor tissue (small glans, proximal or redo cases) and catheter blockage or bladder spasm.

Prevention

- A **barrier layer** (dartos flap, or **tunica vaginalis flap** in proximal or redo repairs) separates the neourethra from the skin.
- Offset suture lines, tension-free glanuloplasty and good drainage.

Evaluation

- Note site, number and size; a coronal fistula suggests glans dehiscence.
- **Rule out distal obstruction** by calibration, urethroscopy and uroflowmetry.

Management

- **Wait at least 6 months** so inflammation settles and tissues mature.
- **Always correct distal obstruction at the same sitting**; otherwise the fistula recurs.

Fistula type	Procedure
Small (<2–3 mm), single	Excise, layered closure, dartos barrier flap , offset skin lines
Large, multiple, thin urethra	Redo urethroplasty
Coronal (glans dehiscence)	Redo glanuloplasty

- Simple closure succeeds in about **80–90%**.

UCF → wait ≥6 months → calibrate/cystoscopy
└ distal obstruction → correct at same sitting
└ small single → excise + layered closure + barrier flap
└ large/multiple/glans dehiscence → redo urethroplasty

Recent advances

- Routine barrier layers and tunica vaginalis flaps in proximal repairs.
- Validated outcome scores (**HOSE (Hypospadias Objective Scoring Evaluation)**, **HOPE (Hypospadias Objective Penile Evaluation)**).

Diagram to draw: cross-section of neourethra, dartos barrier and skin with offset suture lines.

Key points to remember:

- Fistula is the commonest complication; under 5–10% after distal TIP, higher after proximal repair.
 - Always exclude and correct distal obstruction.
 - Wait at least 6 months, then do a layered closure with a barrier flap and non-overlapping suture lines.
 - A coronal fistula means glans dehiscence and needs redo glanuloplasty.
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IV-3b. Silodosin

Type: SAQ (10 marks)

Definition

- Silodosin is a **highly uroselective α 1A-adrenoceptor antagonist**, used mainly for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).

Pharmacology

- Selectivity α 1A : α 1B is about **162 : 1**. Because vascular α 1B receptors are spared, **orthostatic hypotension is minimal**; strong α 1A blockade in the seminal tract causes **ejaculatory dysfunction**.
- Taken **with a meal**; metabolised by **CYP3A4**.

Dose and contraindications

Item	Detail
Dose	8 mg once daily (OD) with a meal
Moderate renal impairment (CrCl (creatinine clearance) 30–50 mL/min)	4 mg daily
Contraindicated	CrCl <30, Child-Pugh C liver disease, strong CYP3A4 inhibitors (e.g. ketoconazole)

Efficacy

- US phase III RCTs (randomised controlled trials)**: improved IPSS (International Prostate Symptom Score) and Qmax within 2–4 days.
- Chapple et al., Eur Urol 2011: non-inferior to tamsulosin, and only silodosin reduced nocturia.**
- It does not shrink the prostate, so add a 5-ARI (5 α -reductase inhibitor) for large glands.

Adverse effects

- Anejaculation (about 28%)** is commonest and reversible; it reflects reduced seminal emission rather than true retrograde flow.
- Orthostatic hypotension <3%.
- IFIS (intraoperative floppy iris syndrome)**: warn before cataract surgery.

Uses

- BPH/LUTS, especially elderly and cardiac patients (low hypotension risk).
- MET (medical expulsive therapy)** for distal stones >5 mm (off-label); the SUSPEND trial (Lancet 2015) showed no overall benefit with tamsulosin.
- Ureteric stent symptoms.

Comparison of α -blockers

Drug	Selectivity	Dose	Orthostatic hypotension	Ejaculatory dysfunction
Silodosin	α 1A (very high)	8 mg OD	Least	Highest (~28%)
Tamsulosin	α 1A = α 1D > α 1B	0.4 mg OD	Low	Moderate (~10%)
Alfuzosin	Clinically uroselective	10 mg OD	Low-moderate	Least

Drug	Selectivity	Dose	Orthostatic hypotension	Ejaculatory dysfunction
Doxazosin, terazosin	Non-selective; titrate	1-8 mg, 1-10 mg	Higher to highest	Low

Recent advances

- Fixed-dose combinations with dutasteride or tadalafil; use in premature ejaculation.

Diagram to draw: α 1 subtypes: α 1A prostate, α 1B vessels, α 1D bladder.

Key points to remember:

- α 1A : α 1B selectivity about 162 : 1; 8 mg OD with a meal, 4 mg if CrCl 30-50; contraindicated if CrCl <30, Child-Pugh C or with strong CYP3A4 inhibitors.
- Anejaculation (~28%) is the commonest side effect; hypotension risk is the lowest of the α -blockers.
- Chapple 2011: non-inferior to tamsulosin and also reduced nocturia.
- Warn about IFIS before cataract surgery.

IV-3c. Sacral neuromodulation

Type: SAQ (10 marks)

See Paper III, Q3a (identical question). Recent advances to add:

- **Rechargeable, miniaturised IPGs (implantable pulse generators)**, such as Axonics r-SNM (sacral neuromodulation) (FDA (Food and Drug Administration) 2019) and InterStim Micro.
 - **Full-body MRI (magnetic resonance imaging)-conditional** systems (1.5 T and 3 T).
 - **Implantable tibial nerve stimulators** (eCoin, Revi).
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IV-3d. Urethral stents

Type: SAQ (10 marks)

Definition

- A tubular endoprosthesis placed endoscopically in the **prostatic or bulbar urethra** to keep it open, for men unfit for definitive surgery.

Classification

Type	Examples	Features
Temporary, non-epithelialising	Memokath (thermo-expandable nickel-titanium), Allium	Not covered by urothelium, so removable
Permanent, epithelialising	UroLume	Covered by urothelium in 3-6 months, so hard to remove; withdrawn
Biodegradable	Experimental	Dissolve

Indications

- BPH (benign prostatic hyperplasia) with retention in men unfit for TURP (transurethral resection of prostate).
- Recurrent short bulbar stricture** in men unfit for urethroplasty.
- DSD (detrusor-sphincter dyssynergia) in spinal cord injury**: reversible alternative to sphincterotomy.

Technique

- Placed under local or regional anaesthesia with endoscopic or fluoroscopic guidance; Memokath expands at about 55 °C and softens when cooled for removal.
- A stricture stent must **not cross the external sphincter**, or incontinence results.

Complications

Complication	Comment
Migration	Commonest with temporary stents
Encrustation and stones	Especially prostatic stents
Hyperplastic tissue ingrowth	UroLume; causes restenosis
Incontinence	Stent crossing the sphincter
Painful erections	Bulbar stents

Results

- Prostatic stents free most unfit men from catheters short term, but many are later removed.
- UroLume worked early, but tissue hyperplasia caused **recurrent obstruction** long term, so it was abandoned.

Place in current guidelines

- AUA (American Urological Association)**: no stent in a man considering urethroplasty, because it makes reconstruction harder.
- EAU (European Association of Urology)**: not routine. Urethroplasty remains standard.

Recent advances

- **Optilume** paclitaxel-coated balloon for strictures ≤ 3 cm: **ROBUST III** showed about 75% vs 27% stricture-free at 6 months compared with DVIU (direct vision internal urethrotomy) or dilation.
- **iTIND** (temporary implantable nitinol device) sits in the prostatic urethra for 5–7 days for BPH.

Diagram to draw: sagittal urethra with prostatic stent and a bulbar stent distal to the sphincter.

Key points to remember:

- Temporary stents (Memokath, Allium) vs permanent epithelialising stents (UroLume, now withdrawn).
 - Indications: unfit BPH patients, unfit bulbar stricture patients, and DSD.
 - Main problems: migration, encrustation and hyperplastic ingrowth.
 - AUA: no stent if urethroplasty is considered; Optilume (ROBUST III) is the recent advance.
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IV-3e. Studer Neobladder

Type: SAQ (10 marks)

Introduction

- An **orthotopic, low-pressure ileal reservoir** anastomosed to the urethra (**Studer, Bern**, mid-1980s).

Principles

- Detubularisation** interrupts peristaltic contractions, keeping pressure low.
- A **spherical shape** (Laplace's law, $P = 2T/r$) gives high capacity at low pressure.
- An **isoperistaltic afferent limb** protects against reflux and allows a tension-free ureteric anastomosis.
- The patient voids by **abdominal straining with pelvic floor relaxation**; continence depends on the intact **rhabdosphincter**.

Technique

- Isolate **54-56 cm of ileum**, ending **25 cm from the ICV (ileocaecal valve)** to preserve bile-salt and B12 absorption.
- Open the **distal 40-44 cm** antimesenterically into a **U**; keep the **proximal 12-14 cm** tubular as the **afferent limb** for both ureters.
- Fold into a **sphere**; join the **most dependent point** to the urethra.
- Capacity grows to **400-500 mL** over months.

Indications and contraindications

- Indications:** radical cystectomy in a motivated patient with a **negative urethral margin**.
- Contraindications:** **impaired renal function** (eGFR (estimated glomerular filtration rate) below about 35-50 mL/min); hepatic dysfunction; positive urethral margin; inability to do **CISC (clean intermittent self-catheterisation)**.

Postoperative care

- Irrigate for mucus; bicarbonate for acidosis; catheter out at about 10-14 days.
- Timed voiding (2 h, then 3-4 h) and a **night alarm** to prevent enuresis and overdistension.

Outcomes

- Daytime continence is about **90%** and night-time continence about **70-80%**.
- Under 10-20% of men need CISC; **women need it more** (hypercontinence).

Complications

- Early:** urine leak, ileus, infection, mucus retention.
- Late:** **hyperchloraemic metabolic acidosis** (ileum absorbs ammonium chloride), **B12 deficiency** (screen from 3-5 years), **ureteroileal stricture (~3-10%)**, stones, incontinence or hypercontinence.

Comparison with other orthotopic neobladders

Neobladder	Key features
Studer	U-shaped pouch with an afferent limb
Hautmann	Longer segment (~60-70 cm) folded into a W ; larger initial capacity

Recent advances

- Intracorporeal robotic Studer neobladder; ERAS (enhanced recovery after surgery).
- Nerve-sparing cystectomy in women to reduce hypercontinence.

Diagram to draw: marked ileal segment, then the spherical pouch with afferent limb.

Key points to remember:

- 54–56 cm of ileum ending 25 cm from the ICV: distal 40–44 cm made into a U, proximal 12–14 cm afferent limb.
 - Detubularisation plus a spherical shape (Laplace) gives a low-pressure, high-capacity reservoir.
 - Continence about 90% by day and 70–80% at night; women more often need CISC.
 - Watch for acidosis and B12 deficiency; avoid if eGFR is below about 35–50 mL/min.
-

IV-3f. Cryotherapy for carcinoma prostate

Type: SAQ (10 marks)

Definition

- **Image-guided transperineal freezing** of the prostate to cause cell death.

Physical principle

- **Joule-Thomson effect:** **argon** gas freezes the needle tip to about -185°C , and **helium** is used for active thawing.
- **Thermocouples** monitor temperature; a **urethral warmer** prevents urethral sloughing.

Generations

Generation	Features
1st	Liquid nitrogen, no imaging; high fistula rates
2nd	TRUS (transrectal ultrasound), urethral warmer
3rd	Argon/helium, 17-gauge needles
4th	Focal or hemi-gland therapy with MRI (magnetic resonance imaging) fusion

Mechanisms of cell death

1. **Intracellular ice** from rapid freezing ruptures cell membranes.
 2. **Vascular injury** causes ischaemic necrosis; this is the main effect at the margins.
 3. Apoptosis at the edge of the ice ball, and an immunological effect.
- Reach -40°C , freeze fast and thaw slowly, and give **two freeze-thaw cycles** (the second enlarges the lethal zone).

Indications

1. **Salvage after radiation failure** (most important): biopsy-proven local recurrence, no metastases, PSA (prostate-specific antigen) preferably $<10\text{ ng/mL}$.
2. **Primary whole-gland:** low- to intermediate-risk disease, men unfit for surgery or radiotherapy, gland under 40–50 mL.
3. **Focal or hemi-gland:** MRI-visible unilateral lesion, in a trial.

Technique

- Transperineal TRUS-guided needles and warmer; freeze, helium thaw, repeat.

Outcomes

- Failure is judged by the **Phoenix definition (nadir + 2 ng/mL)**.
- **Donnelly et al., Cancer 2010:** **non-inferior** to external beam radiotherapy for biochemical progression, but sexual function was worse.

Complications

Complication	Primary	Salvage
ED (erectile dysfunction)	~80-90%	Very high
Incontinence	~1-5%	~5-20%
Rectourethral fistula	<0.5-1%	~1-3%

Comparison with other ablative energies

- HIFU (high-intensity focused ultrasound) heats transrectally; cryotherapy is transperineal and treats anterior tumours.

Guideline positions

- **EAU (European Association of Urology):** primary or focal cryotherapy **only within trials or registries**; salvage is an option.
- **AUA (American Urological Association):** whole-gland cryosurgery may be considered.

Recent advances

- MRI-TRUS fusion focal cryotherapy; PSMA (prostate-specific membrane antigen) PET to select salvage patients.

Diagram to draw: sagittal prostate with ice ball, -40°C isotherm inside the 0°C edge.

Key points to remember:

- Argon freezes and helium thaws (Joule-Thomson); target -40°C with two freeze-thaw cycles.
 - Cell death: intracellular ice, vascular ischaemia (main), apoptosis and an immunological effect.
 - Main role is salvage after radiation failure; ED about 80–90%; rectourethral fistula under 1% in primary treatment, higher in salvage.
 - EAU: primary or focal cryotherapy only within trials or registries; Phoenix definition = nadir + 2.
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Rapid Revision Sheet - Paper IV

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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IV-3a. Etiology and management of hypospadias fistula (SAQ)

- Fistula is the commonest complication; under 5–10% after distal TIP, higher after proximal repair.
- Always exclude and correct distal obstruction.
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- A coronal fistula means glans dehiscence and needs redo glanuloplasty.

IV-3b. Silodosin (SAQ)

- $\alpha 1A : \alpha 1B$ selectivity about 162 : 1; 8 mg OD with a meal, 4 mg if CrCl 30–50; contraindicated if CrCl <30 , Child-Pugh C or with strong CYP3A4 inhibitors.
- Anejaculation ($\sim 28\%$) is the commonest side effect; hypotension risk is the lowest of the α -blockers.
- Chapple 2011: non-inferior to tamsulosin and also reduced nocturia.
- Warn about IFIS before cataract surgery.

IV-3d. Urethral stents (SAQ)

- Temporary stents (Memokath, Allium) vs permanent epithelialising stents (UroLume, now withdrawn).
- Indications: unfit BPH patients, unfit bulbar stricture patients, and DSD.
- Main problems: migration, encrustation and hyperplastic ingrowth.
- AUA: no stent if urethroplasty is considered; Optilume (ROBUST III) is the recent advance.

IV-3e. Studer Neobladder (SAQ)

- 54–56 cm of ileum ending 25 cm from the ICV: distal 40–44 cm made into a U, proximal 12–14 cm afferent limb.

- Detubularisation plus a spherical shape (Laplace) gives a low-pressure, high-capacity reservoir.
- Continence about 90% by day and 70–80% at night; women more often need CISC.
- Watch for acidosis and B12 deficiency; avoid if eGFR is below about 35–50 mL/min.

IV-3f. Cryotherapy for carcinoma prostate (SAQ)

- Argon freezes and helium thaws (Joule–Thomson); target -40°C with two freeze-thaw cycles.
- Cell death: intracellular ice, vascular ischaemia (main), apoptosis and an immunological effect.
- Main role is salvage after radiation failure; ED about 80–90%; rectourethral fistula under 1% in primary treatment, higher in salvage.
- EAU: primary or focal cryotherapy only within trials or registries; Phoenix definition = nadir + 2.